PORTALDOT HACKER HOUSE 2026 · ECOSYSTEM DAPPS

AssayGate

Guarded RWA escrow for critical-minerals offtake — on Portaldot

ERC-721 assay certificate • CFO-capped agent tranches • freeze on grade dispute

✦ Cap 20,000 oUSD • Escrow 100,000 oUSD

ASSAY CERTIFICATE

CO-LUAL-2026-014

Mineral	**Cobalt hydroxide**
Grade	**21.50% Co**
Weight	**24,000 kg**
Origin	**Kolwezi, DRC**
Escrow	**100,000 oUSD**
Tranche cap	**20,000 oUSD**

⚠ FREEZE ON DISPUTE

github.com/icohangar-ops/assaygate

THE PROBLEM

Miners ship first.
Then they wait — and argue.

Provisional payment is hostage

Junior miners deliver concentrate under offtake contracts, then wait while the buyer holds most of the payment pending final assay.

Grade is a dispute, not a number

Seller lab says 21.50% Co. Buyer lab says 19.10%. Weight, moisture and sampling become leverage to delay or discount payment.

Agents make it riskier

An automated payment agent without hard limits can move too much, too fast — with no CFO cap and no instant freeze when evidence changes.

The result: working-capital squeeze for the seller, counterparty risk for the buyer, and no shared, tamper-evident record either side can act on.

THE SOLUTION

One certificate. One escrow. Hard limits on every draw.

- **AssayCertificate (ERC-721)** — shipment id, mine, mineral, origin, grade (bps), dry weight and assay URI, minted by a lab attester.
- **AssayEscrow (ERC-20 oUSD)** — buyer funds provisional payment; the seller is the only payee.
- **CFO cap + agent allowlist** — only allowlisted agents release, and never above the tranche cap.
- **Dispute freeze** — CFO or arbiter freezes instantly; every release reverts until cleared.

DEMO LOT

CO-LUAL-2026-014

Seller	**Lualaba Concentrate SPRL**
Buyer	**Northshore Metals BV**
Funded	**100,000 oUSD**
Released in demo	**15,000 oUSD**
Over-cap attempt	**25,000 — rejected**

HOW IT WORKS

From assay to payment — guarded at each step

STEP 1

Mint

Lab attester mints the ERC-721 assay certificate: grade, weight, origin and assay URI on-chain.

STEP 2

Fund

CFO opens escrow and sets the 20,000 oUSD cap. Buyer funds 100,000 oUSD. Seller is sole payee.

STEP 3

Agent release

Agent previews first: `RELEASE` at 15,000 under cap; `REJECT` at 25,000 — chain reverts `ExceedsTrancheCap`.

STEP 4

Dispute freeze

Buyer re-assay 19.10% vs 21.50%: arbiter freezes. Agent returns `HOLD`; releases revert `FrozenByDispute`.

The agent reads before it sends — and the contract enforces the cap even if the agent is compromised.

WHY PORTALDOT

Built for Portaldot's EVM + WASM future

- **EVM today:** Solidity 0.8.24 contracts (ERC-721 + escrow) deploy via Portaldot's pallet-revive EVM path.
- **WASM roadmap:** Portaldot's dual-runtime design leaves room for native ink!/WASM modules beside the EVM escrow.
- **Low-cost settlement:** frequent small tranche releases need cheap, fast finality — not mainnet-Ethereum fees.

Portaldot V3 testnet (configured & probed)

Chain ID `420420777` · RPC `test-evm-rpc-v3.portaldot.dev`

Read-only `eth_chainId` verified 17 Oct 2026 → `0x190f1ca9`

Gas: tPOTv3 (14 decimals) · Deploy script: `npm run deploy:portaldot`

Honest status: no public faucet yet — deploy lands as soon as tPOTv3 reaches the deployer (organiser allocation / Discord ticket).

DEMO EVIDENCE

It runs. The tests prove the guards.

```
AssayGate — 8 passing (425ms)
✔ mints certificate: grade, weight, origin
✔ funds escrow against the certificate
✔ agent release under CFO cap succeeds
✔ release over CFO cap reverts
✔ arbiter dispute freezes escrow
✔ freeze blocks agent release
✔ non-allowlisted caller cannot release
✔ CFO unfreeze → later release succeeds
```

▶ 3-minute demo video

youtu.be/fZz8EEL_O_g

Mint → fund 100,000 → release 15,000 → 25,000 rejected → dispute freeze → agent HOLD.

One-command proof

`npm install && npm test && npm run demo` — ends `DEMO OK`, no faucet or wallet needed.

MARKET & BUSINESS

Critical minerals need payment infrastructure

Jr.

Junior miners & concentrators — cobalt, lithium, copper and tin suppliers squeezed by provisional-payment terms.

B2B

Offtake buyers & traders — who need assay-linked release conditions instead of manual email-and-bank workflows.

CFO

Finance teams — caps, allowlists and freeze rights mirror real treasury controls.

Business model (proposed): per-shipment SaaS fee + escrow basis-point fee, sold with Cubiczan CFO advisory for mining and trading clients. Built by a finance operator, not a generic payments team.

ROADMAP

Testnet before Nov 5.
Then mainnet, then more minerals.

NOW → NOV 5

Portaldot testnet

Claim tPOTv3, run `deploy:portaldot`, publish contract addresses + deploy tx in the BUIDL.

NEXT

Stronger controls

Cumulative CFO budget per agent, umpire-assay unfreeze path, multi-attester assays.

THEN

Mainnet pilot

One real cobalt/copper corridor, audited contracts, stablecoin settlement beside oUSD demo token.

SCALE

More minerals

Lithium, copper, tin templates; buyer/trader dashboards; lender-ready shipment history.

TEAM

AssayGate

Shyam Desigan — solo builder

Founder, **Cubiczan** — CFO advisory. Finance-controls design (caps, allowlists, freeze rights) translated directly into contract rules. Product decisions, offtake workflow and submission are his; implementation was AI-assisted and disclosed in the repo.

Repo → github.com/icohangar-ops/assaygate

Demo: youtu.be/fZz8EEL_O_g · Contracts MIT, unaudited, hackathon MVP — no real offtake funds custodied.

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